MILITARY SPECIFICATION

SEAL, RING

1. SCOPE

1.1 Scope. This specification covers one type of seal, ring.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

ZZ-R-765 - Rubber, Silicone: Low-and High-Temperature and Tear Resistant.

STANDARDS

FEDERAL

FED-STD-601 - Rubber, Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items.
MIL-STD-1168 - Lot Numbering of Ammunition.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

C13-12-9 - Seal, Ring.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:57Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Material. The material cited on Drawing C13-12-9 shall meet all requirements of ZZ-R-765 with the exceptions listed herein.

3.1.1 Physical properties.

(a) Test slab (unaged).

Hardness, $^{+5}_{-2}$, Shore -A- durometer	80
Tensile strength, minimum psi	950
Ultimate elongation, percent	100 to 150
Compression set, maximum, percent (22 hours at 350°F)	15
Brittle point, minimum, °F	-100
Tensile stress at 90% elongation	750 to 950
Tear resistance, minimum, pounds per inch	75
Specific gravity 25/4°F	1.42 to 1.48

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:57Z
Check the source to verify that this is the current version before use.

(b) End Item.

Weight of seal ring, grams, ± 0.5	7.0
Ultimate elongation, percent	100 to 150
Tensile strength, minimum, pounds	25
Specific gravity 25/4°C	1.42 to 1.48

3.1.2 Color. The color of the silicone rubber shall be natural white.

3.2 Preproduction. Prior to the start of regular production, a preproduction sample of seal rings shall be produced in accordance with this specification for examination and test (see 4.3.1).

3.3 Workmanship. The seal rings shall be free of all defects as listed in MIL-STD-407. There shall be no tears, splits, porosity, nonfills, flow cracks, holes or blistors, when examined under no less than 10X magnafication. In addition, the seal ring shall show no signs of tearing or separation when turned inside out and rolled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:57Z
Check the source to verify that this is the current version before use.

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of five seal rings from each mold cavity shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production. In addition, six test slabs measuring 6 by 6 by 1/8 inches shall be provided for Government examination and testing.

4.3.2 Inspection procedure.

4.3.2.1 For examination. All preproduction sample seal rings shall be examined in accordance with the classification of defects and MIL-STD-105.

4.3.2.2 For destructive tests. After successfully passing the examination specified in 4.3.2.1, sample seal rings shall be tested in accordance with 4.4.4.1.

4.3.3 Acceptance/rejection criteria. The preproduction samples shall meet all the requirements of this specification to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. Contained in any one interfix number of seal rings, no more than one interfix number from one manufacturer of the following is permitted:

(a) One batch (see 6.3) of raw material.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Sampling shall be conducted in accordance with FED-STD-601, Groups of Sampling Plans for Test, table II, group letter A.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample seal rings shall be examined in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For destructive tests. Sample seal rings shall be tested in accordance with 4.4.4.1 and FED-STD-601.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:57Z
Check the source to verify that this is the current version before use.

4.4.3.3 Classification of defects.

(a) Seal, ring (Dwg C13-12-9).

Categories		Defects
Critical:		None defined
Major:	AQL 0.10 percent defective	
101		Workmanship (3.3)
	AQL 1.0 percent defective	
102		O.D. incorrect
103		I.D. incorrect
104		Height incorrect
105		Surfaces not flash-free (Note 3 on drawing)

4.4.4 Tests.

4.4.4.1 Seal rings. Seal rings shall be tested in accordance with FED-STD-601 as follows:

Property	Method	Specimens (quantity, form and dimensions)
Test Slab		
Hardness	3021	3021
Tensile strength	4111	4111 (die III)
Ult elongation	4121	4111 (die III)
Compression set	3311	3311
Brittle point	5311	5311
Tensile stress @90% elong	4131	4131
Tear resistance	4211	4211 (die B)
Specific gravity	14011	14011
End Item		
Weight	----	----
Tensile strength	4111	4111 actual end item
Ult elongation	4121	4111 actual end item
Specific gravity	14011	14011 actual end item

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:57Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging, level C. Each seal ring shall be preserved and packaged to afford adequate protection against deterioration and physical damage from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Seal rings preserved and packaged as specified in 5.1 shall be packed in uniform quantities and in such a manner as to assure carrier acceptance and safe delivery at first destination for immediate use. Containers shall be in accordance with Uniform Freight Classification Rules, or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The seal ring covered herein is a component of the M176 launcher and grenade.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Material. Dow-Corning Silastic 748U has been successfully used in the manufacture of the seal ring. This information is provided as an aid to procurement.

6.5 Supersession data. This specification supersedes Edgewood Arsenal Purchase Description 196-131-900, dated 29 May 1968.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:57Z
Check the source to verify that this is the current version before use.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1330-A232

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:57Z
Check the source to verify that this is the current version before use.